

Data Visualization Final Project

Growth in Popularity of America's National Parks, 1904 - 2024

Observable Notebook: <https://observablehq.com/d/d2731cc18bcfo8e1>

Introduction

In 2024, the number of visits to national parks in the United States reached an unprecedented level, totaling nearly 100 million visits across all 63 national parks in 2024 alone. The recent rise in national park visitation is a dramatic contrast to the obscurity of America's national parks during the early 1900s. In 1904, less than 10 national parks had been established in the United States, attracting a total of about 120,000 annual visits. So how has the attraction to U.S. national parks grown from the early 1900s to today? The objectives of this project are to uncover meaningful trends in national park visitation from 1904 to 2024, and to create a visualization that is able to effectively address the following questions:

- Which national parks have grown the most in recreational visits from 1904 to 2024?
- What historical events have significantly affected park visitation?
- Which time periods saw the most growth across all parks?

The chosen visualization approach aims to answer the questions above through the use of a spike map, where each vertical spike is plotted at the geographic location of a national park. The choice to use spikes on a map is important because compared to other visual encodings such as areas of circles, spikes have the advantage of offering representations of magnitude using length, making them easier to accurately quantify. This is the main rationale for using spikes to quantify annual visitation data for each park, where the length of a spike dynamically adjusts with respect to the number of recreational visits to the park across time. Through an interactive horizontal slider spanning the period from 1904 to 2024, the map can be updated based on the year of the slider handle, allowing users to visualize annual visitation data for each park over time. This temporal map-based visualization approach is important not only because it clarifies park-level, regional-level, and national-level visitation trends, but also because it allows users to temporally connect visitation trends with real-world events, thereby addressing all three questions the project aims to answer.

Dataset Description

For this project, the dataset is a combination of two datasets featuring annual visitation data provided by the National Park Service. Both datasets were derived using visitation query builders provided by the National Park Service's Visitor Use Statistics website. One of the two datasets contains annual visitation data from 1979 to 2024 for each of the 63 U.S. national parks. The other dataset contains annual visitation data from 1904 to 1978 for the same set of national parks. Both datasets, as well as the combined dataset, contain the following four features for each record:

- Park Name: the name of the national park.
- Unit Code: a four-letter code uniquely identifying the national park.
- Year: the year of the record.
- Recreation Visits: the estimated number of visits during the year of the record.

The combined dataset also contains the following features in addition to the four features above:

- Latitude: the latitude of the park.
- Longitude: the longitude of the park.
- Acres: the estimated size of the park in acres during the year of the record.

Latitude and longitude coordinates for each park were obtained using the Google Maps interface. Annual acreages for each park come from the National Park Service through their annual acreage reports and through historical electronic resources provided by their park history program. Sources documenting the history of the National Park Service, such as the historical electronic publications found on npshistory.com, were also used to obtain and confirm historical acreage data.

To build the dataset, the following preprocessing steps were involved.

1. Drop duplicate rows from the 1904 to 1978 dataset.

When exporting datasets built using the visitation query builders provided by the National Park Service, I noticed that there were occasional duplicates of rows. This issue was solved using Python, where I used the `drop_duplicates()` function from the pandas library to remove all duplicate rows.

2. Combine the 1904 to 1978 dataset with the 1979 to 2024 dataset.

With Python, I used the `concat` function from the pandas library to combine rows of the 1904 to 1978 dataset with rows of the 1979 to 2024 dataset. This operation resulted in a combined dataset that contains annual visitation data from 1904 to 2024 for each national park.

3. Merge the combined 1904 to 2024 dataset with park coordinate data from Google Maps.

An interesting and particularly useful feature of the combined dataset is that national parks are labeled with “unit codes”, which are four-letter codes with capital letters that uniquely identify each national park. The fact that the dataset contains unit codes for each park makes merging datasets with the same sets of national parks straightforward through the use of the unit code column as the key. So to append longitude and latitude data for each park to the combined dataset, I created another dataset with the same set of parks from the visitation dataset, and added longitude and latitude columns manually populated with park coordinates from Google Maps. Then by using the unit code column as the key, I was able to use the `merge` function from the pandas library to merge the park coordinate dataset with the visitation dataset.

In the visualization, park coordinate data is used in conjunction with two GeoJSON files used to draw boundaries for states and unincorporated U.S. territories. One of these files is used to draw the 50 states, and comes from Roger Veciana i Rovira’s `d3-composite-projections` Github repository. The other GeoJSON file is used to draw the five unincorporated U.S. territories, and comes from the United States Census Bureau.

4. Add rows for missing years.

An important goal of the project is to visualize the gradual addition of parks into the National Park System over time. However, one of the patterns I discovered in the data was that parks with less visits in general were more likely to have time periods with missing visitation data. I gathered that for several of these parks, the first year during which the park has visitation data is not necessarily the same as the year the park was added to the National Park System. This makes it challenging to tell from the data alone the exact year when a park was authorized to become a national park.

To solve this issue, I had to look up additional information about when each park was authorized by Congress to become a part of the National Park System. To incorporate this information into the dataset built up to this point, I manually added rows for each missing year between a park's authorization year and 2024. In each missing row, I added a placeholder of zero recreational visits, which is not misleading because this value is not intended to be displayed in the visualization. The visualization should instead display "Unknown visits" for any park with a placeholder of zero visits in a given year. At first, I thought these placeholders would conflict with another pattern I noticed in the data, which was that several parks seem to have had zero recreational visits during certain years or time periods. Upon further research, it turns out that these numbers are actually misleading. Several parks have zero visits in certain years not because the visitation count was actually zero, but because these parks decided not to count visitation during these years. So by displaying that a park had an unknown number of visits in a given year when the number of visits is zero, the visualization avoids displaying misleading data.

5. Add acreage data for each park.

Because the dataset does not include acreage data for each park, I had to utilize additional resources such as official annual acreage reports provided by the National Park Service. These reports were useful as a starting point, but were limited in that they only covered years between 1997 and 2025. To find older acreage data, I explored sites such as npshistory.com, where I was able to find electronic publications that often reveal historical acreage statistics for U.S. national parks. Adding this information to the dataset involved extending the dataset to include an "Acres" column, and manually populating the column with historical annual acreage data derived from npshistory.com.

6. Adjust misleading park-level data.

As I was exploring the data, a pattern I observed was that several parks seemed to experience extreme shifts in visitation across short periods of time. For example, recreational visits to the North Cascades National Park dropped significantly from 456,444 to 22,796 between the years 1990 and 1991. After further research into the possible causes of this dramatic drop in visitation, I discovered a statement by deputy chief of visitor services Katy Hooper explaining that between 1990 and 1991, the park stopped counting visitors to two of its most popular sites, Ross Lake and Lake Chelan. Fortunately, the National Park Service provides annual visitation data for both of these sites. So to avoid misleading users, this data was added to the annual visitation counts for the North Cascades National Park for years after the apparent drop in visitation.

For time periods when the visitation of a park significantly drops and rises back to the level before the drop, further research was performed to confirm whether these shifts are logically sound. If the shifts were found to be misleading due to changes in counting procedures, linear interpolation was performed using neighboring visitation counts to estimate the misleading values.

While I did notice other dramatic differences in visitation between years for certain parks, these sudden differences were more aligned with what I had discovered online in terms of events happening in the years when the large differences occurred. For example, the extreme spike in visitation of Gateway Arch National Park between 1965 and 1966 makes sense given that the construction of the park was completed in 1965.

Design Rationale

In developing the visualization, many design decisions were made based on considerations of their effectiveness, how well they improve the experience for the user, or how much they contribute to the goals of the project. The sections below detail these design decisions and the reasonings behind them.

Spikes and Circles

The reasoning behind using spikes to quantify park visits is that it is easier for users to compare varying lengths of spikes with each other compared to other alternatives such as varying areas of circles. To further assist users in comparing lengths of spikes that are on top of each other, I designed the map such that hovering over a park highlights it by changing the color of the associated spike and circle to black, and by doubling the stroke width of both the spike and the circle. The decision to highlight parks in this manner is effective, as it allows users to clearly visualize the length of a spike by differentiating it among several spikes located in the same vicinity. The reason why I added circles at the bottom of each spike is to make it easier to visualize and hover over parks with low numbers of visits. Without the added circles, spikes for parks with low visitation appear as thin horizontal lines, making them harder to visualize and hover over.

Because the goal of the visualization is to draw attention to the overall trends in visitation over time, highlighting spikes that move in the same direction as the national trend in visitation is an effective way to emphasize the national trend. One way to do this is to delay the movement of spikes that do not move in the same direction as the national trend, which allows users to see the important trend-following spikes first. This technique is based on the law of common fate, which describes how elements that move together are perceived as one group. Because users perceive trend-following spikes as one group, the national trend is simplified, helping users gain insights into visitation trends more efficiently.

Map of the United States

An important rationale for visualizing national parks on a map is that a map is able to show geospatial relationships among national parks. These relationships are important as they allow users to easily identify visitation trends in specific geographic areas. Because several U.S. national parks are located on unincorporated territories such as the Virgin Islands, the visualization uses a composite Albers projection of the United States that includes Albers projections of all five unincorporated U.S. territories. Since the unincorporated territories of the U.S. are relatively distant from the lower 48 states, a composite projection is an effective solution because it minimizes distortion of distant regions by utilizing separate projections for each territory. This allows them to be positioned closer to the map in insets, which are

enclosed by composition borders to indicate their geographic separation from the lower 48 states. The main reason for using the Albers projection is that it produces more accurate representations of countries that are larger in extent from east to west, such as the United States.

Bar Chart

One of the interesting patterns I noticed during exploratory data analysis was that several parks experienced extreme fluctuations in visitation across short periods of time. In some years such as 2015 and 2020, several parks had experienced extreme fluctuations that align in the same direction. To highlight these patterns, the visualization features a bar chart displaying the total recreational visits across all parks for the current year. This chart is effective at highlighting moments when several parks experience extreme, parallel fluctuations in visitation because the bar dramatically shifts during these periods, drawing users' attention to extreme visitation trends. Additionally, the bar features animated transitions between years, further drawing users' attention to extreme bar movements.

Line Chart

A line chart above the year slider was added to more clearly visualize the overall growth of U.S. national park visitation from 1904 to 2024. While the animated bar chart is also used to visualize this growth, it is not enough to visualize the overall trend across time because the bar only shows total visits for individual years rather than for all years at once. A line chart is effective for displaying visitation growth across time because it offers accurate annual visitation data through a clear and easy-to-understand interface, which helps readers clearly and effectively identify key visitation trends across time.

To help users connect real-world events with trends in national park visitation, the line chart features several textual annotations, each describing a major event that preceded or happened during years when many parks experienced significant visitation shifts in the same direction. When the slider handle is positioned under an annotation, the annotated text becomes darker. The reasoning behind this design decision is not only to help readers be able to read important events above the handle more clearly, but to also draw readers' attention to these events, which helps readers retain annotated information while promoting active engagement with the visualization.

Interactive Elements

On the map, tooltips assist with accurately quantifying visitation data by displaying these values when a reader hovers over a park. This functionality is intended to give readers the ability to easily derive accurate visitation data among spikes that may be harder to visually compare, such as those that are spatially distant from each other. Tooltips are also necessary for displaying additional park information not represented on the map, such as acreage data. The year slider is necessary for allowing readers to filter visitation data by year, and to also explore how the National Park System has grown in visitation and number of parks over time. An interactive year slider is effective because it gives readers full control over the speed at which the years progress. This allows readers to gain insights across different time periods with varying rates of visitation growth. As users adjust the slider, circles that appear are temporarily darkened in order to clearly indicate the year when they were added into the National Park System, as well as to give users an overall sense of how many parks were added during each year.

The numeric filters at the top-right corner of the visualization are important for giving readers the ability to filter parks by visits and acres. Numeric input fields were chosen because they provide readers the functionality to filter national parks using precise values. Performing this task is much harder using

sliders, especially on smaller screens where the width of the slider may be too small to allow for fine-grained control. Below the filters is a checkbox that gives users the ability to filter parks that have unknown visits. This functionality is necessary when filtering parks by visits leaves parks with unknown visits remaining on the map, which may need to also be filtered to display only parks with data that truly reside within the numeric filters. The visualization also includes a “Reset Filters” button to help readers revert back to the original values of the input fields with a single click, which is important for offloading the manual task of resetting each field to help users focus more on exploring the data.

A tutorial featuring guiding annotations was implemented in order to guide users through each interactive element. This tutorial is necessary for revealing to users the various ways of interacting with the visualization that are possible, but may not be immediately obvious at first glance. As for other textual elements, an overarching title and historical information below the title were added to the top of the design. These textual elements are important for giving context to the visualization, which is important for users who may not be as familiar with the history of national parks in the United States.

Consideration of Alternative Designs

As for visual alternatives, I considered incorporating an animated year slider that would update the map automatically as the slider moves across time. I considered this alternative design choice because it would allow users to devote more mental energy towards understanding overall trends in visitation over time without having to manually adjust the slider. The problem with an animated year slider is that years may move too quickly to offer insights during recent years when there is extreme fluctuation, such as 2015 and 2020. Although an option to adjust the speed of the slider could be implemented, this would defeat the purpose of the design choice because the user would need to devote mental energy increasing the animation speed during periods when year-to-year shifts in visitation are smaller, and decreasing the animation speed during periods when shifts in visitation are larger.

Another design alternative I considered implementing is a tree graph that contains nodes for each park, and grows in number of nodes as the user moves the year slider from 1904 to 2024. To represent visitation growth for each park in the tree, branches that connect to each park would extend as the years progress. But because parks may become visually cluttered at the root of the tree when branches are smaller in length, I thought it would be best to evenly space parks along a central vertical line, which would grow as the number of parks grows. At this point, the design alternative was simply a bar chart with branches as “bars”.

I thought about the idea of implementing a bar chart containing spikes for each park. This would be effective because readers would be able to clearly compare lengths of spikes since all spikes would align along the same axis. But ultimately, I thought this design alternative would not be as satisfying as a map-based approach featuring the locations of each park, which has the benefit of displaying visitation trends in specific geographic areas.

Development Process

I worked alone on this project, which means that instead of distributing tasks among members of a team, tasks were distributed among the following temporal phases of the project, aligned with the timeline of the data visualization course:

- Week 10: Dataset Research and Initial Data Acquisition, Formation of Visualization Ideas
- Week 11: Data Analysis and Preprocessing, Additional Research
- Weeks 12 to 14: Design Development

Week 10: Dataset Research and Initial Data Acquisition, Formation of Visualization Ideas

During week 10 of the data visualization course, I devoted around 25 hours to researching and selecting datasets to use for the visualization, and brainstorming preliminary design ideas and alternatives for visualizing the data.

I allocated about 15 hours to researching and selecting a datasets to use for the visualization. In order to survey a wide range of potential datasets to use, I found it useful to use FiveThirtyEight's Data Is Plural newsletter, which features weekly articles featuring datasets covering a wide range of unique and important fields. After about three hours of exploring a number of different datasets, I discovered an article from June of 2019 that mentioned national park data. And because visits to national parks have often made me curious about their vast cultural and natural history, the topic inspired me to research further into the corresponding data source linked in the article. About five hours were dedicated to finding clean and useful datasets from the linked source. I eventually discovered the NPS Stats Portal, which features several interactive query builders that offer downloadable CSV files containing national park visitation data. Around seven hours were devoted to becoming familiar with these interactive tools to derive datasets covering annual park visitation data from 1904 to 2024.

After building the two datasets, I spent around 10 hours brainstorming preliminary design ideas for visualizing the data. Half of these hours were spent researching Observable projects for inspiration, which led me to come up with alternative design ideas such as animated sliders and interactive tree graphs. Once I settled on major design elements, I spent the other half of these hours researching examples of how these elements have been implemented using d3, and whether I would successfully be able to implement these features given the limited amount of time to complete the project.

Week 11: Data Analysis and Preprocessing, Additional Research

During week 11, I devoted around 40 hours to analyzing and preprocessing the data, which also involved conducting additional research to augment the data in order to make the chosen visualization approach possible. One of the first steps during this phase was to analyze both datasets for possible errors or inconsistencies. This step took around 30 minutes since both datasets were fairly clean upon initial inspection. The only adjustment I had to make was to drop duplicate rows from the 1904 to 1978 dataset. To perform this operation, as well as derive the combined dataset, I spent about 30 minutes writing a Python program to remove duplicate rows and to combine both datasets, using functions from the pandas library such as the `drop_duplicates()` and `concat()`.

After cleaning and combining the data, the next step was to augment the data with geographic coordinates for each park, using coordinate data from Google Maps. This was a very tedious process because there were 63 parks that I needed to look up geographic coordinates for, which were to be added to a separate dataset in order to use Python to merge the coordinates with the main dataset. In total, the process of populating park coordinates into a separate dataset, and merging this data with the original data, took around 10 hours.

The next step was to add rows for missing years between each park's authorization year and 2024. This step was not feasible to do programmatically because authorization years varied from park to park, meaning that it was not possible to simply add rows for all missing years between 1904 and 2024. The process of gathering the authorization year of each park was straightforward because the National Park Service website features an article on park anniversaries, which lists the authorization years of all national parks in one place.¹ The majority of time on this step was spent manually adding rows for missing years across all parks, which took about 10 hours. This process involved using the Numbers app to identify which rows were missing, and using the gathered authorization years from the National Park Service to insert rows for each missing year after each park's authorization date.

14 more hours were allocated for augmenting the dataset with annual acreage data for each park. About 10 of these hours were spent researching historical acreages due to the fact that much of this information cannot be conveniently found in one place. Gathering historical acreage data for each of the 63 parks involved conducting investigative research into the historical timeline of each park, using websites with historical park information such as nps.gov and npshistory.com. Fortunately, sizes of national parks have not frequently changed from year to year, which meant that the only information that needed to be gathered were the years when major park expansions or contractions took place. After collecting this information, four hours were dedicated to using the Numbers app to manually piece together an "Acres" column featuring estimated annual acreages for each park from 1904 to 2024.

The last major task during this phase was to identify and correct potentially misleading park-level visitation data. This task involved a combination of exploring the data to find unusual patterns in the data, such as extreme visitation shifts during short periods of time, and researching whether these patterns are logically valid with respect to the events that happened around the time of these patterns. This identification process took about three hours. After identifying the potentially misleading data, I spent about two hours correcting the data by collecting additional National Park Service visitation data for parks that may have removed areas from counting procedures, and by using methods such as linear interpolation for when there is a clear trend surrounding the misleading values.

Weeks 12 to 14: Design Development

The remaining three weeks were devoted to implementing the chosen visualization approach using d3. The majority of the design was completed during weeks 12 and 13, with week 14 being devoted to implementing suggestions from student critiques. In total, implementing the visualization took around 75 hours.

During week 12, around two hours were allocated to becoming familiar with Roger Veciana i Rovira's d3-composite-projections module, referencing instructions in the Github repository where the module sources from. In the repository, I also happened to come across a convenient GeoJSON file of the 50 states, allowing these states to be projected onto the map. Figuring out how to project this GeoJSON

¹ <https://www.nps.gov/subjects/npscelebrates/park-anniversaries.htm>.

data onto the map using the `geoAlbersUsaTerritories` projection from the `d3` composite projections module took an additional two hours. After an hour was spent finding GeoJSON data from the United States Census Bureau for the remaining unincorporated U.S. territories, 30 minutes were spent analyzing this data to identify indices of the territories that allow them to be added to the map without redundantly adding the other 50 states already drawn. Projecting these territories onto the map using the same projection for the 50 states took an additional 30 minutes.

After completing the map layout, the next step was to add spikes onto the map. On Observable, Mike Bostock has created a public notebook that greatly helped with implementing spikes on the map using triangular paths. With this notebook, I was able to study the code for the spike map and implement spikes reflecting park visitation data in around three hours. The notebook was also a helpful reference for understanding how to implement the legend, which took about one hour to implement. Because the spikes needed to reflect visitation data for specific years, the year slider needed to be implemented next. John Walley's simple SVG slider library was used to implement this feature, and I allocated around one hour to becoming familiar with the library. While the process of adding the slider to the visualization was straightforward, figuring out how to update spikes on the map to reflect park visitation for the current year was much more challenging, taking around 14 hours to implement. This implementation is based on the idea of adjusting the visibility of spikes based on whether the years they are associated with correspond to the slider year. After implementing this logic, the slider began to act as a visual filter for displaying visitation data corresponding to the current year.

Adding circles to the bases of spikes took about one hour to implement, as the code for adding them to the design is fairly similar to the code for adding spikes. The ability for spikes to become highlighted when readers hover over them was more challenging to implement, taking about 10 hours. The solution for this implementation was to add class attributes to each spike and each circle, where each class corresponds to the unit code of the associated park. Classes allow specific spikes and circles to be selected, making it possible to update style attributes of these elements when a user hovers over them.

Week 12 involved adding several textual elements to the design. The title and text underneath it were surprisingly challenging to implement due to the fact that typical SVG elements do not have word-wrap functionality. To solve this issue, I decided to use `foreignObject` elements that add html containers with word-wrap functionality to display text. Learning how to implement `foreignObject` elements and incorporating them into the visualization took a total of around five hours. Learning how `foreignObject` elements work was useful for implementing annotations above the year slider, as these annotations also require word-wrap functionality and need to use `foreignObject` elements. Annotations above the year slider took about six hours to implement, involving the creation of vertical ticks for each annotation, implementing the annotations with `foreignObject` elements, and trial-and-error adjustment of each annotation's position in the layout. `html` containers are also used for tooltips, which took close to five hours to implement due to having to research how to format large numbers to include commas, and having to adjust the dimensions of tooltips for certain parks with longer park names.

During week 13, the bar chart was added to the visualization above the map. Implementing this bar chart took about three hours, involving the addition of a simple rectangle with labels for the title and the horizontal axis. The chart uses `d3` transitions to interpolate the width of the bar when transitioning between years. To calculate total visits across all parks by year, the data was transformed using `d3`'s `rollup` function to aggregate the totals. Week 13 also saw the implementation of spike transitions. Between years, I wanted transitions for spikes that moved in a different direction from the direction of the national trend to be longer by a few milliseconds. However, implementing this animation was more challenging than I anticipated because `d3` cannot easily handle transitions between years when both years differ in the

number of parks on the map. Realizing this underlying issue took about an hour, and to solve the issue, I spent another hour in Observable implementing code that adds placeholder rows to the dataset such that every park has the same number of rows. After solving this issue, I was able to spend the next two hours implementing the desired animated transitions between years.

The last part of week 13 was spent implementing the filters at the top-right corner of the visualization. These were also challenging to implement because of the fact that they use html elements that are generally harder to work with in d3. I spent about an hour adding the numeric filters to the visualization, but implementing their functionality took nearly four hours. The main challenge I faced was that event handlers for each input field did not update the map properly when changing the values of these fields. After significant trial and error, I realized that the values typed into these fields were not being interpreted as integers. By applying the `parseInt` function to the input values, the actual values typed into these fields could be interpreted properly and the map could be updated. Implementing the unknown visits check box was much more straightforward, as the feature does not rely on user input. So adding the checkbox to the visualization and implementing its functionality took close to 30 minutes.

Week 14 was spent implementing suggestions from student critiques. I implemented six student suggestions, and allocated around 11 hours for implementing them using d3. The first three suggestions I implemented were the revision of the title to include “Growth in Popularity”, the darkened annotation texts when dragging the slider handle underneath them, and a reset button for the filters in the top-right corner. These features took a total of two hours to implement as many of them were simple adjustments or involved simple implementation logic. I implemented the line chart next, which took about two hours to implement. Adding this feature involved a combination of figuring out how to position the line chart in the layout such that it does not clutter the view, figuring out how to use d3’s built-in line generator to draw the path on the layout, and adding y-axis labels. Because adding this chart meant that event annotations also needed to be repositioned, I also spent nearly one hour adjusting the positions of these annotations within the chart.

Implementing the tutorial for interacting with the visualization was tricky to implement due to the logic involved with aligning user interactions with the order in which labels should appear. The implementation took nearly five hours and is based on toggling the visibility of five `foreignObject` elements containing annotated text, based on user interactions that have and have not been performed. An hour was allocated to implementing the temporary darkening of park circles when they appear on the map. This implementation is based on increasing the opacities of circles that appear after a user has performed an action that updates the map, and then using d3 transitions to gradually adjust the opacities back to their original values.

Feedback Incorporation

At the beginning of week 14, I received three student critiques from Philip, Yetong, and Tina regarding an earlier design of the visualization. I was selective in incorporating the suggestions I thought would most improve the visualization, and those that I thought would be the most feasible to implement over the span of a week. Given that the visualization at the beginning of week 14 had many features already, incorporating every suggestion would not have been the most effective use of time because too many additional features would clutter the layout, potentially making the visualization harder for users to understand and interact with.

Philip mentioned that the design includes several effective aspects, including the temporal map-based approach, the use of spikes, the circles at the bases of the spikes, and the contextual annotations highlighting real-world events. Regarding Philip's constructive feedback, the main takeaway I gathered was that there needed to be more guiding visuals or text when users are first presented with the visualization. To address this, I added additional guiding labels that together act as a tutorial for how to interact with the visualization. These labels improve the visualization not only because they give readers a clear idea of how to interact with the visualization, but also because they encourage engagement with the design by explaining to readers several ways of how to interactively reveal additional data insights. Without the tutorial, users may not know how to interactively explore the data, leading to some users exploring less of the data and gaining less insights.

Yetong described how the visualization contains several elements that help to clearly capture how America's National Park System has grown in visitation and number of parks over time. For example, Yetong mentioned that the interactive year slider and spikes are effective at highlighting visitation trends because they allow users to easily compare annual visitation data across all parks. Yetong also thought the historical milestones beneath the slider are effective at providing meaningful context to help users connect real-world events with shifts in visitation. An interesting suggestion that Yetong gave was to integrate the year slider with a line chart that displays overall visitation trends across all parks. To incorporate this suggestion, I added a line chart above the year slider that displays the overall growth of total annual recreation visits across all parks from 1904 to 2024. This feature improved the visualization by providing users with a clear and intuitive trendline of total annual visitation across all parks, which helps users gauge overall visitation trends at a glance. Without the line chart, these trends would have been harder to identify because the map and bar chart only show data pertaining to the slider year, and do not display simultaneous data useful for quickly identifying visitation trends across many years.

In Tina's critique, Tina mentioned that the vertical spikes were easy to understand and well-proportioned to each other in terms of length. She also wrote positively of interactive features such as the numeric input fields to filter parks by acres, but wished there was a reset button for these features. I addressed this feedback by adding a "Reset Filters" button below the four input fields, which allows users to restore the original values of the filters, as well as the original state of the check box for filtering parks with unknown visits. The button improves the visualization by helping users easily return to a more familiar state when filter values have dramatically altered the map. When the map significantly differs from its normal state after filtering parks by visits and acres, users may not know how to continue with the analysis, which makes the button useful for guiding users back to a point where they can analyze the data further.

Tina also suggested revising the title to more clearly describe the growth in popularity of national parks. By changing "Growth" to "Growth in Popularity", the visualization more clearly communicates its focus on visitation growth rather than on acreage growth. I also implemented Tina's suggestion to darken event annotation texts when the slider handle is underneath them. This functionality improves the visualization by clearly highlighting important events that happened at specific points in time, improving memory retention of annotated information while also promoting user engagement with the design. The last important suggestion from Tina that I implemented was the temporary darkening of park circles when they appear on the map. This suggestion improved the visualization by making it easier to determine years in which more (or less) parks were added to the National Park System. Thus, the suggestion gives users the ability to derive trends in the overall demand for national parks across time.

Final Thoughts

To conclude, analyzing the development of America's National Park System from 1904 to 2024 was a challenging, but educationally-rewarding experience that greatly improved my skills and knowledge in data analysis, d3, and design development. Additionally, the project gave me the opportunity to learn deeply about the history of America's national parks in terms of visitation and acreage. During data analysis and preprocessing, I learned how to efficiently clean and wrangle data using functions from the Python pandas library such as `drop_duplicates()`, `concat()`, and `merge()`. The analysis phase also taught the importance of conducting research to verify unusual patterns identified in the data. It was discovered that some extreme shifts in visitation did not actually happen, and were instead the result of changes in visitation counting procedures.

During the design development phase, I improved my skills in analyzing the effectiveness of visual encodings by brainstorming and considering design alternatives. By devoting time to this step of the project, issues identified in the alternative designs were avoided, potentially saving time and effort that would have been spent on these designs. During the design implementation phase, I gained a deeper understanding of d3 as a powerful and versatile tool for developing novel interactive visualizations. The versatility of d3 allowed me to explore how to represent data using unique visual shapes such as vertical spikes. In exploring how to use the d3 library, I also learned how to implement interactive features and animated transitions to further increase user engagement with the design. Reading student feedback presented me with opportunities to improve my visualization further, as well as improve my programming abilities using d3. And by writing critiques for other students' projects, I further improved my design analysis skills by analyzing the effectiveness of a wide range of design and visual encoding choices and how they can be improved.

In future projects, I might consider spending more time analyzing the usability of potential datasets in order to reduce the amount of preprocessing required to align the data with project goals. While the datasets offered by the National Park Service did not require a significant amount of cleaning, I had to perform a significant amount of data wrangling because I intended to visualize additional information not included in the datasets, such as location data. In many cases though, data wrangling is hard to avoid when a single dataset that contains all of the information needed for a project does not exist. However, future projects can potentially minimize the amount of data wrangling required by allocating more time for surveying a wider range of datasets before the preprocessing phase. Another potential improvement in future projects would be to study a wider range of functionality in the d3 library, which would help in efficiently identifying the most effective d3 functions or techniques for a variety of visualization tasks. As I create more visualizations using d3, I imagine my knowledge of the functionality and versatility of the library will grow over time, contributing towards my ability to efficiently create d3 visualizations that closely align with the goals of future projects.

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